

HISTOLOGY

- Concerning the *teeth and associated tissues*,
 - Dentin is more calcified than enamel.
 - Pulp is dense connective tissue.
 - Enamel is replaced in the root by cementum.
 - Odontoblasts make enamel.
 - Periodontal membrane is loose connective tissue.
- Which of the following types of *lingual papillae* lack *taste buds*?
 - circumvallate
 - filiform
 - foliate
 - fungiform
 - none
- Regarding the *stomach*, which statement is *true*?
 - Is able to withstand the acidic environment because it is lined by stratified squamous keratinized epithelium.
 - Has its gastric glands best developed in the pylorus than the other regions.
 - forms temporary folds called plicae circulares when the stomach is empty.
 - Has its muscularis externa made by skeletal muscle fibres.
 - None of the above.
- Pernicious anemia can result from insufficient intrinsic factor secretion by which of the following cells?
 - Paneth cells
 - enterocytes
 - ameloblasts
 - oxyntic (parietal) cells
 - zymogenic (chief) cells
- Concerning the *stomach*, which statement is *true*?
 - It is specifically responsible for the absorption of vitamin B₁₂ and the reabsorption of conjugated bile salts.
 - Goblet cells are more abundant to produce mucus as well as appreciable amounts of lysozyme.
 - Hydrochloric acid and intrinsic factors are secreted by enteroendocrine cells of cardia and pylorus.
 - Parietal cells also called oxyntic cells contain mucinogen granules and pepsinogen.
 - The folds of the empty stomach are called rugae.
- Which one of the following marks best the *ileum* from the *duodenum* and *jejunum*?
 - Long and broad leaf like villi.
 - Absence of teniae coli.
 - Peyer's patches.
 - Plicae circulares.
 - Well developed intestinal glands.
- Which of the following organs characteristically contains the most *goblet cells* in the mucosa covering its *villi*?
 - duodenum
 - ileum
 - colon
 - cecum
 - stomach
- Vermiform appendix*,
 - has few lymphoid nodules in the lamina propria and submucosa.
 - is a blind-ended evagination of the ileum.
 - has relatively shallow intestinal glands.
 - longitudinal smooth muscle gathers into its teniae coli.
 - contains short villi.

9. With regard to the *small intestine*, which statement is *true*?
- The muscularis externa in the upper third of duodenum and jejunum is composed of striated muscle.
 - Crypts of Lieberkühn are lymphatic capillaries [lacteals] within a villus of the small intestine.
 - Microvilli that seen on the columnar epithelial cells in the small intestine form the brush border or striated border.
 - The plicae circulares are projections with a core of mucosa.
 - Villi are finger like projections of submucosa.
10. Diarrhea may result if which of the following organs fails to carry out its primary role in *absorbing water from the feces*?
- anal canal
 - colon
 - cecum
 - small intestine
 - stomach
11. With regard to the *large intestine*, which statement is *false*?
- Paneth cells of the submucosal glands house bacteria that produce vitamin B₁₂ and vitamin K.
 - Lacks villi and contains a disproportionately large number of goblet cells.
 - The lamina propria has many macrophages, plasma cells, eosinophils, and lymphoid nodules.
 - The muscularis externa has three thick bands of muscle called the tenia coli.
 - The large intestine is shorter than the small intestine but has a wide lumen.
12. *Hepatic sinusoids*,
- are bordered directly by hepatocytes.
 - contents flow toward the portal triads.
 - are surrounded by the space of Disse.
 - contents empty into the canal of Herring.
 - lumen is entirely sealed by junctional complexes.
13. Which is *not* the *function of the liver*?
- Detoxification/removal of drugs and alcohol.
 - Storage of calcium and bile.
 - Storage of glycogen, fat soluble vitamins, iron and cholesterol.
 - Production of plasma proteins and lipoproteins.
 - Fetal RBC production
14. With regard to the *pancreas*, which statement is *false*?
- Although it is a mixed exocrine and endocrine gland, the majority of the secretory cells are exocrine cells.
 - The myoepithelial cells of the striated ducts deliver bicarbonate ions into the primary secretion.
 - The exocrine portion is a compound tubuloacinar gland, similar in structure to the parotid gland.
 - Islets of Langerhans are incompletely delimited from the acini by Schwann cells.
 - It is a lobulated retroperitoneal gland whose islet cells are compact mass of epithelial cells not cords or plates.
15. Concerning the *GI tract*, select the statement which is *not correct*.
- The mucosa of the gallbladder is made of simple columnar epithelium.
 - The masticatory mucosa is located on the hard palate and gingiva.
 - The adventitia consists of only a single layer of epithelial cells and underlying connective

tissue.

- d. The muscularis externa consists of Auerbach's plexus.
- e. Brünner's glands in submucosa of the duodenum produce mucin containing alkaline fluid and the polypeptide hormone urogastrone.

16. In the *digestive system*, select the statement which is *false*.

- a) Parotid glands produce purely serous secretion in the adult.
- b) Paneth cell secretes lysozyme.
- c) the exocrine pancreas contain centroacinar cells.
- d) the hepatic portal vein carry blood rich in bilirubin.
- e) cholecystokinin stimulates the relaxation of smooth muscle in the gallbladder.

17. Which organ in the GIT has a *mucosa lined by nonkeratinized stratified squamous epithelium*?

- a. Esophagus b. Stomach c. Small intestine d. Large intestine e. Rectum

18. What type of cells are found in the olfactory mucosa?

- a. Basal cells b. Brush cells c. Olfactory cells d. Sustentacular cells
- e. All of the above

19. Which of the following *components of the respiratory tract* wall increase(s) in amount from trachea to alveoli?

- a) cilia b) elastic fibers c) smooth muscle d) cartilage e) goblet cells

20. What type of epithelium is found in the *respiratory mucosa* of man?

- a. Non-ciliated pseudostratified columnar epithelium with goblet cells
- b. Ciliated pseudostratified columnar epithelium with goblet cells c. Transitional epithelium
- d. Simple columnar epithelium e. Stratified squamous epithelium

21. Which cell is a *respiratory macrophage*?

- a. Kupffer cell b. Histocyte c. Dust cell or alveolar macrophage
- d. Langerhans cell e. Microglia

22. *Bronchioles*:

- a) Have well developed cartilages b) Are lined by stratified squamous epithelium
- c) Have no muscle d) Possess Clara cells e) Have mucous producing glands

23. The *visceral pleura*,

- a) can be described as an adventitia. b) is devoid of lymphatic vessels.
- c) includes a layer of simple squamous epithelium. d) lacks elastic fibers.
- e) covers the outer wall of the pleural cavity.

24. *Pulmonary surfactant*,

- a) is secreted by type I pneumocytes. b) may promote the growth of bacteria.
- c) consists of a monolayer of phospholipids overlying an aqueous hypophase.
- d) prevents alveolar collapse by increasing surface tension.

e) excessive production in the lungs of premature infants causes respiratory distress syndrome.

25. *Pneumocytes type I*:

- a) are found in large numbers in the terminal bronchioles.
- b) Could give rise to pneumocytes type II.
- c) Produce surfactant.
- d) Have large round centrally located nuclei.
- e) Line more surface area of the alveolar wall than do the alveolocytes type II.

26. Select the *incorrect* statement in the *respiratory system*.

- a) The vestibule of the nose is lined by stratified squamous epithelium.
- b) The paranasal sinuses are lined by ciliated pseudostratified epithelium with goblet cells.
- c) Larynx is partly lined by nonkeratinized stratified squamous epithelium.
- d) A smoker's respiratory epithelium typically develops more goblet cells.
- e) The lamina propria contains mucous glands at the level of the alveolar ducts.

27. Which of the following is not a *function of the kidney*?

- a. Erythropoietin production
- b. Vitamin D modification
- c. Acid base balance
- d. Aldosterone production
- e. Renin production

28. What vessels arise from the *efferent arterioles*?

- a. Afferent arteriole
- b. Efferent arteriole
- c. Capillary tuft
- d. Peritubular capillaries
- e. Vasa recta

29. Which of the following is *not* the function of *mesangial cells*?

- a. synthesis of extracellular matrix.
- b. endocytosis and contractility.
- c. maintenance of acid-base balance in the blood.
- d. structural support to the glomerulus.
- e. increase blood flow and the effective surface area available for filtration.

30. Select the *correct* statement pertaining the *morphology of the kidney*.

- a. The distal convoluted tubules exhibit brush borders and apical canaliculi.
- b. The renal pyramid and its associated cortex is referred to as the renal lobe.
- c. Macula densa cells are located in the urinary pole of the nephrons.
- d. Renin is secreted from the visceral layer of Bowman's capsule referred to as area cribrosa.
- e. Interstitial cells in the renal medulla secrete erythropoietin through the filtration slits.

31. In the *urinary system*, which statement is *true*?

- a) The proximal convoluted tubule lacks a brush border.
- b) The cortical collecting tubules are located primarily at the apex of the renal pyramids and are lined by pseudostratified ciliated columnar cells.
- c) Extraglomerular mesangial cells are tall, narrow, closely packed epithelial cells of the distal tubule.
- d) Endothelial cells of glomerular capillaries are of non-fenestrated varieties.
- d) None of the above.

32. *Testis* has the following coverings *except*:

- a. tunica adventitia b. tunica vasculosa c. tunica albuginea d. tunica vaginalis
e. All of them

33. In the *testis*, which statement is *true*?

- a. Interstitial cells (of Leydig) produce testicular fluid.
b. Sertoli cells are directly involved in spermatogenesis.
c. The secondary spermatocytes are the largest cells of the spermatogenetic lineage.
d. The tunica vaginalis is thickened on the posterior surface to form the mediastinum testis.
e. During spermiogenesis, the spermatids quickly undergo the first meiotic division to give spermatozoa.

34. In the *male reproductive system*, which structure is *not* lined by *pseudostratified epithelium*?

- a. vas deferens. b. seminal vesicles. c. epididymis
d. seminiferous tubules e. None of them

35. Which of the following statements is *false* in the *male reproductive system*?

- a. Sertoli cells coagulate semen and form a vaginal plug.
b. The subcutaneous tissue in penis is devoid of fat.
c. The secretion of seminal vesicle is particularly rich in fructose.
d. The largest male accessory gland, the prostate gland, is composed of compound tubuloalveolar glands found in fibromuscular [fibroelastic] stroma.
e. Cowper's glands [bulbourethral glands] are homologous to the greater vestibular glands in the female.

36. When does a *secondary oocyte* complete its second meiotic division to become a mature ovum?

- a) at ovulation b) before ovulation c) at fertilization
d) at puberty e) before birth

60. Primary oocyte completes its first meiotic division at the time of ovulation

37. Which of the following is *true* of the *ampulla* of the *uterine tube* but not of its *isthmus*?

- a) Contains large mucosal folds that are extensively branched.
b) Contains smooth muscle in its walls. c) Is covered by serosa derived from peritoneum.
d) Contains ciliated epithelial cells. e) Contains mucus-secreting peg cells.

38. Which of the following statements is *true* about the *female reproductive system*?

- a. Mucosa of the uterus is commonly called perimetrium.
b. Fornix of vagina is lined by stratified squamous epithelium.
c. The life span of corpus luteum of pregnancy is about 10 lunar months.
d. Primary oocyte completes its first meiotic division at the time of fertilization.
e. Nonciliated secretory cells [peg cells] present in the lining epithelium of uterine tube synthesise and store glycogen.

39. Which of the following statements is *true* in the *female reproductive system*?

- a. The cells of theca interna secrete progesterone.

- b. Germinal (surface) epithelium covering the ovary gives rise to ova.
- c. Glassy membrane is the last structure to disappear in an atretic follicle.
- d. The alkaline environment of vagina prevents growth of microorganisms.
- e. The basic pH in the vaginal lumen is secretion of HCl by the tubuloalveolar glands of the posterior vaginal fornix.

40. In the *female reproductive system*, which statement is *true*?

- a) The peg cells of the uterus secrete a nutrient-rich medium that nourishes the spermatozoa and preimplantation embryo.
- b) The basal layer of the endometrium is the thick smooth muscle tunic of the uterus.
- c) The inferior portion of the cervix, which is continuous with the lining of the vagina, is covered by stratified squamous keratinized epithelium.
- d) The epithelium of the vagina contains glycogen.
- e) Mammary glands are simple tubuloacinar glands.

47. The endometrical layer that is shed during each menstrual cycle is called stratum functionalis.

48. Granulosa cells that are adherent to the zona pellucida form corona radiata.

53. Corpus luteum is derived from theca interna and membrana granulosa of the ovarian follicle.

55. Mammary gland is a modified apocrine sweat gland.

56. Primordial follicle is identified in section by the presence of

- a. zona pellucida
- b. single layer of flattened follicular cells
- c. theca folliculi
- d. antrum folliculi
- e. cumulus oophorus

61. Secretory phase of endometrium is characterized by the presence of

- a. oedematous endometrium
- b. tortuous uterine glands
- c. secretion rich in glycogen
- d. all of the above

63. Lactating mammary gland presents all of the following features except that

- a. alveoli are large and branched
- b. alveoli contain secretory material
- c. connective tissue septae are thick and well developed
- d. intralobular ducts are few

64. Granulosa lutein cells of corpus luteum secrete progesterone

66. Germinal epithelium covering the ovary gives rise to ova. FALSE.

67. Glassy membrane is the last structure to disappear in an atretic follicle. TRUE.

68. Zona pellucida is derived from oocyte only. FALSE.

69. The cells of theca interna secrete progesterone. FALSE.

70. During pregnancy the myometrium undergoes hypertrophy. TRUE.

71. Straight arteries supply the superficial layer of the endometrium. FALSE.

72. Coiled arteries are under the influence of ovarian hormones. TRUE.

73. Proliferative phase of uterus coincides with the development of ovarian follicles. TRUE.

75. Colostrum gives passive immunity to the newborn. TRUE.

_____ 11. In the *ovaries*, which statement is true?

- a) The capsule contains tunica albuginea of dense irregular collagenous connective tissue.
- b) The primary follicles are dependent on FSH for their development.
- c) Corpus albicans is a temporary endocrine gland whose formation depends upon LH surge.
- d) Atretic follicles are found in the ovarian medulla.
- e) Theca interna cells manufacture androgens.

54. In the female reproductive system, which of the following statements is false?

- a. Ovulation is triggered by a dramatic increase in LH.
- b. During a pregnancy, hCG maintains the corpus luteum.
- c. Proliferative phase of uterus coincides with the development of ovarian follicles.
- d. In the ovaries, the medulla harbors most of the oocyte-containing ovarian follicles.
- e. During pregnancy the endometrium grows extensively by hypertrophy and hyperplasia.

55. Which of the following is true of the uterine tube's ampulla but not of its isthmus?

- a) contains mucus-secreting peg cells
- b) contains ciliated epithelial cells
- c) is covered by serosa derived from the peritoneum
- d) contains smooth muscle in its walls
- e) contains large mucosal folds that are extensively branched

56. Which of the following statements is false in the female reproductive system?

- a. Mammary gland is a modified apocrine sweat gland.
- b. The vagina lacks glands throughout its length and is lubricated by secretions from cervical glands.
- c. Lactating mammary gland secretes progesterone.
- d. Cervix does not participate in the endometrial (menstrual) phases.
- e. None of the above.

3. Which stage of the follicle is arrested in prophase?

- a. Primordial follicle
- b. Primary follicle
- c. Secondary follicle
- d. Mature follicle
- e. Graffian follicle

4. What is the cavity within a secondary follicle?

- a. Graffian follicle
- b. Theca folliculi
- c. Granulosa cells
- d. Zona pellucida
- e. Antrum

During the first week to 10 days, FSH is the main hormone which stimulates the growth of the follicles. Ovulation is triggered by a dramatic increase in LH.

During a pregnancy, HCG is the hormone which maintains the corpus luteum.

7. Which stage of the follicle is marked by the surrounding of flattened (squamous) follicular cells becoming cuboidal?

- a. Primordial follicle
- b. Primary follicle
- c. Secondary follicle
- d. Mature follicle
- e. Graffian follicle

The connective tissue layer around the primary follicle is the theca folliculi.

The acidophilic glycoprotein coat surrounding the oocyte is the zona pellucida.

A cavity within the follicle is the antrum.

Underneath the germinal epithelium of the ovary is connective tissue called the tunica albuginea.

During a pregnancy, hCG is the hormone which maintains the corpus luteum.

12. Which follicular stage is also called an antral follicle?

- a. Primordial follicle b. Primary follicle c. Secondary follicle
- d. Mature follicle e. Graffian follicle

2. What is the acidophilic glycoprotein coat which surrounds the oocyte?

- a. Graffian follicle b. Theca folliculi c. Granulosa cells d. Zona pellucida
- e. Antrum

5. Which stage of the follicle is characterized by a surrounding of flattened (squamous) follicular cells?

- a. Primordial follicle b. Primary follicle c. Secondary follicle
- d. Mature follicle e. Graffian follicle

8.

What is the connective tissue layer around the primary follicle?

- a. Graffian follicle b. Theca folliculi c. Granulosa cells
- d. Zona pellucid e. Antrum

10. What is the connective tissue which surrounds the ovary?

- a. Follicle b. Germinal epithelium c. Medulla d. Tunica albuginea
- e. Cortex

_____ 1. Concerning *oogenesis*, which statement is true?

- a) All oogonia become primary oocytes during ovulation.
- b) Primary oocytes begin the first meiotic division before birth.
- c) Oogonia grow and give rise to secondary oocytes by mitotic division.
- d) Primary oocytes undergo cytodifferentiation to form three polar bodies prior to ovulation.
- e) The difference in sex chromosome complement of oocytes forms the basis of primary sex determination.

_____ 2. Concerning *spermatogenesis*, one statement is false.

- a) Type A spermatogonia undergo mitosis to provide a continuous supply of stem cells throughout the reproductive life of the male.
- b) Primary spermatocytes in males begin the first meiotic division before birth, but completion of prophase does not occur until adolescence.
- c) The primary spermatocytes are the largest germ cells in the seminiferous tubules.
- d) No primary spermatocytes form after birth.
- e) None of the above.

_____ 3. With regard to *spermatogenesis*, which statement is false?

- a. spermatids are transformed into spermatozoa by spermiogenesis.
- b. The entire process of spermatogenesis takes approximately 72 hours.

- c. Spermatogenesis is regulated by LH production by the pituitary.
- d. Spermatogenesis begins at puberty.
- e. Spermatogonia begin to increase in number at puberty.

_____ 5. Regarding the *sperm transport*, which statement is false?

- a) Vesiculase coagulates some of the semen and forms a vaginal plug.
- b) Fructose in the semen, secreted by the seminal glands, is an energy source for the sperms.
- c) Spermatozoa are nonmotile during storage in the epididymis, but become motile in the ejaculate.
- d) Prostaglandins in the semen stimulate uterine motility.
- e) None of the above.

_____ 7. Which statement is true about *female reproductive cycles*?

- a) LH stimulates the development of ovarian follicles and the production of estrogen by its follicular cells.
- b) ovulation is triggered by a surge of FSH production.
- c) if an oocyte is fertilized, the corpus luteum involutes and degenerates.
- d) ischemic phase occurs when the oocyte is fertilized.
- e) None of the above.

_____ 8. Concerning the *female reproductive system*,

- a) the usual site of fertilization is the ampulla of the uterine tube.
- b) the endometrium has a wall made up of a very thick smooth muscle layer.
- c) the basal layer of endometrium is sloughed off during each menstruation.
- d) the perimetrium is a trumpet-shaped opening towards the peritoneal cavity.
- e) the vaginal glands are sloughed off and discarded with each menstrual flow during pregnancy.

_____ 9. Which statement is false?

- a) The basal layer of the endometrium is sloughed off during menstruation.
- b. If a sperm penetrates the secondary oocyte, the second meiotic division is completed.
- c. The number of oocytes that ovulate is greatly reduced in women who take oral contraceptives.
- d. At ovulation, the nucleus of the secondary oocyte begins the second meiotic division, but progresses only to metaphase.
- e. Removal of the corpus luteum of pregnancy before the 4th month usually leads to abortion.

_____ 10. With respect to the *zona pellucida*, which statement is false?

- a. is composed of sulfated glycoproteins.
- b. prevents polyspermy
- c. prevents premature implantation
- d. functions as a porous filter
- e. arises by delamination from the embryoblast.

_____ 11. *Liquor folliculi*,

- a) stimulates the myoepithelial cells of the mature follicle to rupture.
- b) acts as a porous filter through which dissolved hormones and enzymes are prevented.
- c) causes the endometrial glands to secrete and prepare the endometrium for implantation.
- d) is composed of cumulus oophorus and polar bodies.
- e) None of the above.

55. Which of the following statements is *true* about the *female reproductive system*?

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- d. The alkaline environment of vagina prevents growth of microorganisms.
- e. The basic pH in the vaginal lumen is secretion of HCl by the tubuloalveolar glands of the vaginal walls.

EMBRYOLOGY

1. The proximal convoluted tubules of the definitive adult kidney are derived from
 - a) ureteric bud
 - b) metanephric vesicle
 - c) metanephric duct
 - d) pronephric tubules
 - e) metanephric tubules
2. Concerning the development of the *urinary system*, which statement is *false*?
 - a) Urinary system begins to develop before the genital system.
 - b) The metanephric diverticulum is the primordium of the ureter.
 - c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
 - d) Differentiation of the nephrons depends upon induction by the collecting tubules.
 - e) The kidneys attain their definitive adult position toward the end of the fetal period.
3. Concerning the *development of the urinary bladder and urethra*, which statement is *false*?
 - a) The female urethra is derived from endoderm of the urogenital sinus.
 - b) In adults, the urachus is represented by the median umbilical ligament.
 - c) The urinary bladder becomes a pelvic organ during the 9th week of development.
 - d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
 - e) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.
4. With regard to the *congenital anomalies of the urinary system*, which statement is *false*?
 - a) Accessory renal arteries are end arteries.
 - b) The “ascent” of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
 - c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.
 - d) Unilateral renal agenesis is more common in males than females.
 - e) None of the above.
5. During the *development of the Urinary System*, which statement is *true*?
 - a) The urorectal septum divides the urachus from the allantois.
 - b) Bilateral renal agenesis is associated with oligohydramnios.
 - c) The podocytes of Bowman’s capsule are derived from endoderm and ectoderm.
 - d) Urachal cyst is a remnant of the cloaca.
 - e) Urine production begins shortly after birth.

The metanephros become functional just before birth.xxxxxxxxxxxxxx

The transitional epithelium lining the ureter is derived from endoderm and neural crest cells.xxxxxx

6.

7.

8. A structure found within the *adult female pelvis* formed from the *gubernaculum* is
 a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament
9. The *labia minora* arise embryologically from
 a) phallus b) labioscrotal swellings c) sinovaginal bulbs
d) urogenital folds e) paramesonephric duct
10. The *uterine tubes* of the adult female are derived embryologically from
 a) mesonephric duct b) mesonephric tubules
c) paramesonephric duct d) paramesonephric tubules e) uterovaginal primordium
11. In the *male*, *failure of urethral folds to fuse* completely results in
a) hypospadias b) epispadias c) cryptorchidism
 d) hydrocele e) congenital inguinal hernia
13. With regard to the development of the *thyroid gland*, which statement is *false*?
 a) The thyroid primordium consists of a solid mass of endodermal cells.
 b) The thyroid gland is connected to the tongue by a narrow tube, the thyroglossal duct
 c) The proximal opening of the thyroglossal duct persists as a small pit in the dorsum the tongue-the foramen cecum.
d) After birth, the levels of fetal thyroid -stimulating hormone and thyroxine begin to increase, reaching adult levels during puberty..
 e) The parafollicular cells (C cells) in the thyroid gland are derived from the ultimopharyngeal bodies.
13. Concerning the development of the *suprarenal gland*, which statement is *true*?
 a) The cortex develops from endoderm and the medulla differentiates from intermediate mesoderm.
 b) The zona reticularis and zona fasciculata are present at birth.
 c) The fetal medulla produces steroid precursors that are used by the placenta for the synthesis of estrogen.
 d) Cells in the fetal cortex do not proliferate; they give rise to the thin, poorly defined zona glomerulosa.
e) None of the above.
12. With regard to the development of the **endocrine system**, which of the following statements is **false**?
a) the oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.
 b) C cells differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of pharyngeal pouches.
 c) the thyroid gland is the first endocrine gland to develop in the embryo.

- d) the pituitary gland is purely ectodermal in origin.
- e) the pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.

13. The epoophoron and paroophoron, which are important as cysts, arise from:

- a) The Mullarian ducts
- b) The paramesonephric ducts
- c) The mesonephric ducts
- d) Leydig hormones
- e) Wolffian hormones

28. Concerning the development of the urinary System, which statement is *false*?

- a) Collecting duct is the adult derivative of the ureteric bud.
- b) Urinary system begins to develop before the genital system
- c) The podocytes of the Bowman's capsule are derived from neural crest cells.
- d) Nephrons are derived from the metanephric mass of intermediate mesoderm.
- e) The kidneys attain their definitive adult position toward the end of the fetal period.

29. During the *development of the kidney systems*, which statement is *true*?

- a) Urine production begins shortly after birth.
- b) At birth, the kidneys have a lobulated appearance.
- c) The transitional epithelium lining the nephrons is derived from neuroectoderm.
- d) The "ascent" of the horseshoe kidney is prevented by the root of the inferior mesenteric artery.
- e) The pronephros are derived from intermediate mesoderm from the upper thoracic to the upper lumbar segments.

30. Concerning the *development of the urinary bladder*, which statement is *false*?

- a) In adults, the urachus is represented by the median umbilical ligament.
- b) The urinary bladder becomes a pelvic organ during the 9th week of development.

- c) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
- d) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.
- e) None of the above.

31. With regard to the ***development of gonads***, which statement is ***false***?

- a) The surface epithelium of the female gonad develops into oogonia.
- b) Sex of an embryo is determined genetically at the time of fertilization.
- c) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
- d) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.
- e) A dense tunica albuginea is the characteristic feature of testicular development in the fetus.

32. Concerning the ***development of the genital system***, one statement is **false**?

- a) Bilateral cryptorchidism results in sterility.
- b) Hypospadias is constantly associated with exstrophy of the urinary bladder.
- c) The fetal testis produces antimüllerian hormone by sertoli cells at 6-7 weeks.
- d) In testicular feminization syndrome, uterus and uterine tubes are generally absent.
- e) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or the labia majora.

33. Concerning the ***development of the genital system***, which statement is ***false***?

- a) Inguinal canals develop in both sexes.
- b) Round ligament of the uterus is formed from the gubernaculum.
- c) Gubernaculum does not play any role in the descent of ovaries.
- d) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
- e) The most common cause of testicular feminization syndrome is lack of androgen receptors.

34. With regard to the **vestigial structures** derived from embryonic **genital ducts**, which statement is **true**?
- a) The prostatic utericle is homologous to the vagina.
 - b) The cranial end of mesonephric duct in males may persist as paradidymis.
 - c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis.
 - d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
 - e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of
the uterine tube may persist as hydatid of Morgagni.

35. With regard to the **development of the male reproductive system**, which statement is **false**?
- a) Failure of urethral folds to fuse completely results in epispadias.
 - b) The seminiferous cords develop into the seminiferous tubules, tubuli recti, and rete testis.
 - c) The Y chromosome carries a gene on its short arm that codes for testes-determining factor.
 - d) Urethral glands and paraurethral glands of the female correspond to the prostate in the male.
 - e) The paramesonephric ducts degenerate except for a small portion at their cranial ends, the appendix testis.

36. During the development of the **female reproductive system**, which statement is **true**?
- a) Round ligament of the uterus is formed from the gubernaculum.
 - b) The labia minora arise embryologically from paramesonephric ducts.
 - c) Primary female sexual differentiation in the fetus is determined by antimüllerian hormone.
 - d) The uterine tubes of the adult female are derived embryologically from uterovaginal primordium.
 - e). None of the above.

Embryology #13 – Urinary System

- 1) The urinary system develops mainly from Intermediate mesoderm.
 - d) Lateral plate mesoderm
- 5) The ureteric bud gives rise to the _____ and the metanephric mesoderm (blastema) gives rise to the _____ .

- a) Nephrons; Collecting system b) Collecting system; Nephrons
- c) Bowman's capsule; Ureter d) Loop of Henle; Renal pelvis
- e) Distal/proximal convoluted tubules; Major/minor calyces

6) As the kidney ascends, it rotates from _____ -facing to _____ -facing (relating to the hilum facing direction).

- a) Anterior; Inferior b) Posterior; Lateral c) Anterior; Lateral
- d) Posterior; Medial e) Anterior; Medial

7) The urorectal septum divides the Anorectal canal from the Urogenital sinus

8) The urachus forms the:

- a) Allantois
- b) Median umbilical ligament
- c) Medial umbilical ligament
- d) Lateral umbilical ligament
- e) Round (umbilical) ligament

9) The urogenital sinus is made from what type of tissue?

- a) Muscle
- b) Nerve
- c) Connective
- d) Epithelial

10) The bladder becomes continuous with the lower end of the mesonephric (Wolffian) duct, which will form the _____ of the bladder.

- a) Urethra
- b) Prostate
- c) Trigone
- d) Posterior wall
- e) Anterior wall

11) At birth, a child has large palpable flank masses and abnormal extremities. The delivery physician is very concerned as she knows this autosomally-dominant disorder can lead to renal failure and the child will likely need a kidney transplant to survive. Which of the following disorders does the child likely have?

- a) Polycystic kidney
- b) Renal agenesis
- c) Pelvic kidney
- d) Bifid ureter
- e) Bifid kidney

12) This disorder, associated with oligohydramnios, occurs when the ureteric bud fails to reach the metanephric mesoderm.

- a) Polycystic kidney
- b) Renal agenesis
- c) Pelvic kidney
- d) Horseshoe kidney
- e) Bifid ureter

13) What artery prevents the horseshoe kidney (metanephric mesodermal fusion) from ascending high into the abdomen?

- a) Celiac trunk
- b) Superior mesenteric artery
- c) Inferior mesenteric artery
- d) Left renal artery
- e) Right renal artery

14) Which of the following disorders would show an increased level of alpha fetoprotein (AFP) during maternal Quad testing/screening?

- a) Renal agenesis
- b) Horseshoe kidney
- c) Bifid ureter
- d) Ectopic ureter
- e) Exstrophy of the bladder

15) A newborn baby presents with fluid draining from the umbilicus onto the skin. Testing of the fluid identifies it as urine. What is the most likely diagnosis?

- a) Urachal cyst
- b) Urachal fistula
- c) Exstrophy of the bladder
- d) Bifid ureter
- e) Horseshoe kidney

Embryology #14 – Reproductive System

1) Testis determining factor (TDF), a transcription factor, is turned on by a(n) _____ chromosome from the _____.

- a) X; egg
- b) Y; egg
- c) X; sperm
- d) Y; sperm

2) The indifferent gonad forms as a pair of longitudinal ridges _____ to the mesonephric ridges.

- a) Medial
- b) Lateral
- c) Superior
- d) Inferior

3) The germ cells arise from the epiblast (week 2) and are in the yolk sac during week 3. At week 5, they migrate through the dorsal mesentery of the _____ and arrive at the genital ridge in week 6.

- a) Foregut
- b) Midgut
- c) Hindgut
- d) Cloaca
- e) Cervical region

4) All of the following contribute to the indifferent gonad EXCEPT:

- a) Ectodermic placodes
- b) Endodermic germ cells
- c) Coelomic epithelium
- d) Mesenchymal cells

5) During development of the ovaries, the primitive sex (medullary) cords _____.

- a) Proliferate
- b) Move laterally
- c) Move medially
- d) Differentiate
- e) Degenerate

6) The gubernaculum is attached to the uterus and as the ovaries descend it becomes the _____ of the uterus.

- a) Canal of Nuck
- b) Ovarian ligament
- c) Round ligament
- d) Ovarian and round ligaments
- e) Gubernacular ligament

7) Development of the testis is induced by the *Sry*-gene, which encodes testis-determining factor (TDF). TDF causes all of the following EXCEPT:

- a) The sex cords continue to proliferate to form the testis
- b) Leydig cells are secreting testosterone thus have very little SER
- c) Towards the hilum, the cords grow into a network that will give rise to the rete testes
- d) During further development, a thick layer (tunica albuginea) separates the cords from the surface epithelium
- e) Sex cords become seminiferous cords that form seminiferous tubules (Sertoli cells), straight tubules, and rete testes

8) Which of the following inhibits the growth of the paramesonephric duct (males)?

- a) Testosterone
- b) Leydig Cells
- c) Dihydrotestosterone
- d) Estrogens
- e) Mullerian substances (MIFs)

9) As the testes descend, they follow the:

- a) Processus vaginalis
- b) Parietal layer of tunica vaginalis
- c) Visceral layer of tunica vaginalis
- d) Caudal genicular ligament
- e) Cryptorchidism

10) The epididymis, ductus deferens, ejaculatory duct, and seminal vesicle all arise from:

- a) The Mullarian ducts
- b) The paramesonephric ducts
- c) The mesonephric ducts
- d) Leydig hormones
- e) Wolffian hormones

11) The uterine tubes, uterus, cervix, and superior part (1/3) of the vagina all arise from:

- a) The Wolffian ducts
- b) The paramesonephric ducts
- c) The mesonephric ducts
- d) Leydig hormones
- e) Mullarian hormones

12) The epoophoron and paroophoron, which are important as cysts may form from these remnants, arise from:

- a) The Mullarian ducts
- b) The paramesonephric ducts
- c) The mesonephric ducts
- d) Leydig hormones
- e) Wolffian hormones

13) The lower portion of the vagina is derived from:

- a) Sinovaginal bulb
- b) Uteral lumen
- c) Hymn
- d) Urogenital sinus
- e) Vaginal plate

14) What structure in the indifferent stage (weeks 3-6) becomes the clitoris?

- a) Urogenital groove
- b) Genital swelling
- c) Urethral fold
- d) Perineum
- e) Genital tubercle

15) What structure in the indifferent stage (weeks 3-6) becomes the phallus?

- a) Urogenital groove
- b) Genital swelling
- c) Urethral fold
- d) Perineum
- e) Genital tubercle

16) Genetic 46XY would be:

- a) Turner's syndrome
- b) Klinefelter syndrome
- c) True hermaphrodite
- d) Male pseudo-hermaphrodite
- e) Female pseudo-hermaphrodite

17) A child is born with a tumor on their adrenal glands. This leads to adrenalyperplasia. They display 46XX phenotypical characteristics. Which of the following is the best description of this patient:

- a) Turner's syndrome
- b) Klinefelter syndrome
- c) True hermaphrodite
- d) Male pseudo-hermaphrodite
- e) Female pseudo-hermaphrodite

18) A patient is born with karyotype 46XY but has no problem with production of testosterone or mullarian inhibiting hormone. Their vagina ends in a blind pouch and testes are lodged in the inguinal canal. Which of the following is the best description of this patient:

- a) Male pseudo-hermaphrodite
- b) Female pseudo-hermaphrodite
- c) Androgen insensitivity syndrome
- d) Turner's syndrome
- e) Klinefelter syndrome

19) A young boy is learning how to use a standup urinal and must hold his penis pointing superiorly due to a birth defect. Which of the following describes his defect?

- a) Bifid penis
- b) Hypospadias
- c) Hydrocele
- d) Cryptorchidism
- e) Bicornis

19) The pyramidal lobe, seen in some individuals, is remnant from what structure?

- a) Tongue
- b) Hyoid bone
- c) Pharyngeal gut
- d) Thyroglossal duct
- e) Foramen cecum

32. Concerning the development of the genital system, which statement is false?

- a) Inguinal canals develop in both sexes.
- b) Sertoli cells constitute most of the seminiferous epithelium in the fetal testis.
- c) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
- d) Gubernaculum does not play any role in the descent of ovaries.
- e) The Y chromosome carries a gene on its short arm that codes for testes-determining factor (TDF).

11. During the development of the testes, which statement is false?

- a) A dense tunica albuginea is the characteristic feature of testicular development in the fetus.
- b) Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester.
- c) The fetal testis produces antimüllerian hormone by sertoli cells at 6-7 weeks
- d) Sertoli cells are derived from the surface epithelium of the testis.
- e)

The labia minora arise embryologically from paramesonephric ducts.fff

The uterine tubes of the adult female are derived embryologically from uterovaginal primordium.fff

The most common cause of testicular feminization syndrome is lack of androgen receptors.ttt

Round ligament of the uterus is formed from the gubernaculum. ttt

The Y chromosome carries a gene on its short arm that codes for testes-determining factor (TDF).ttt

88. The most common cause of female pseudointersexuality is

- a) a 46, XO genotype
- b) a 47, XXY genotype
- c) lack of androgen receptors
- d) congenital adrenal hyperplasia
- e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

- _____ 89. The most common cause of male pseudointersexuality is
- a) a 46, XO genotype
 - b) a 47, XXY genotype
 - c) lack of androgen receptors
 - d) congenital adrenal hyperplasia
 - e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

21. During the urinary system,

- a) the metanephros begins to form at week 5.
- b) if the ureteric bud undergoes division prematurely, duplication of the ureter and renal pelvis occurs.
- c) the urinary bladder develops from the upper end of the urogenital sinus.
- d) the ureter develops from the ureteric bud .
- e) the ureteric bud is a diverticulum from the mesonephric duct.
- f) the podocytes of Bowman's capsule develop from the metanephric vesicles.
- a) the transitional epithelium lining the entire urinary bladder is of endodermal origin.

22. Trigone on the posterior wall of the urinary bladder is formed by,

- a) incorporation of the lower end of the mesonephric ducts.
- b) incorporation of the lower end of the pronephric ducts.
- c) incorporation of the metanephric blastema.
- d) incorporation of the mesonephric tubules.
- e) incorporation of the pronephric tubules.

_____20. Skeletal muscle may occur in the muscularis externa at the junction between which of the following pairs of organs?

- a) pharynx and esophagus
- b) esophagus and stomach
- c) stomach and duodenum
- d) duodenum and jejunum
- e) ileum and cecum

_____20. Myoepithelial cells,

- a) are usually found outside the acinar basal lamina
- b) can not contract
- c) contain actin and myosin in their cytoplasm
- d) are found around the bile canaliculi.
- e) are responsible for emptying the gallbladder.

_____20. Which of the following glands lacks serous acini but contains serous demilunes?

- a) liver
- b) parotid gland
- c) pancreas
- d) sublingual gland
- e) submandibular gland.

_____20. Regarding the gastric glands:

- a) Are best developed in the pylorus than in the other regions of stomach.
- b) Has no submucosa.

- c) Those in the pyloric region, have short pit, but long glands.
 d) Are simple branched in type.
 e) Those in the cardiac region have more type of secretory cells than those of fundus.
- _____20. The centroacinar cells of pancreas are:
 a) Larger in size than the secretory cells. b) More eosinophilic than the secretory cells.
 c) Are found in the middle of the Islet of Langerhans. d) More basophilic than the secretory cells.
 e) Make the beginning of the intercalated ducts.

- _____20. Which of the following is not a component of the *portal triad of the liver*?
 a) Branch of hepatic artery b) Branch of portal vein
 c) Bile duct d) Central vein of liver e) None of the above

- _____74. Which of the following is a factor produced or activated in the dermis that stimulates calcium uptake in the digestive tract?
 a) calcitonin b) intrinsic factor c) melanin d) vitamin A e) vitamin D

- _____57. Permanent circular folds (plicae circulares) is a characteristic structural feature of the walls of which of the following organs?
 a) anal canal b) colon c) cecum d) small intestine e) stomach
2. The absorptive cells of the small intestine,
 a) are also called enteroendocrine cells. b) many microvilli cover their basal surfaces.
 c) absorb lipids by active transport. d) use absorbed lipids to synthesize triglycerides.
 e) undergo mitosis at the tips of the villi and are sloughed into the crypts.

- _____33. With regard to the oral cavity, which statement is true?
 a. Sebaceous and eccrine sweat glands are prominent in the transitional [vermillion] zone of a lip.
 b. The internal region of the lip is covered by a stratified squamous keratinized epithelium.
 c. In the submucosa of the cheeks are minor salivary glands and occasional sebaceous glands (Fordyce spots).
 d. Pseudostratified columnar ciliated epithelium covers the oral surface of the soft palate.
 e. The palatal raphe is a median fold that joins the internal aspect of each lip to the gum.

- _____34. Concerning the histology of the teeth, which statement is true?
 a. The highly calcified matrix of enamel is composed of type III collagen fibrils.
 b. The organic matrix of enamel is secreted by odontoblasts.
 c. Dentin is a calcified connective tissue that is harder than bone.
 d. Ameloblasts are tall, columnar epithelial cells that line the pulp chamber.
 e. Like bone, cementum is perforated by Sharpey's fibres and neurovascular elements.

- _____35. With regard to the tongue, which statement is false?
 a. The filiform papillae are the smallest and most abundant papillae on the dorsum of the tongue.
 b. Filiform papillae are the only papillae on the tongue which do not contain taste buds.
 c. Foliate papillae are well developed in man and are seen on the edges of the tongue arranged in a "V" shape.

- d. Taste buds are found on circumvallate, fungiform, and foliate papillae.
- e. Fungiform papillae are mushroom shaped papillae on the tongue which contain taste buds.

5. *Mucosal glands of the pylorus,*

- a) resemble those in the body of the stomach.
- b) secrete mainly mucus.
- c) secrete more pepsinogen than do fundic glands.
- d) have shallower pits than those in the fundic region.
- e) contain abundant parietal cells in their basal half.

_____ 42. Concerning the Kupffer cells, which statement is false?

- a. uptake, store, and release of retinoids.
- b. are found on the luminal surface of the endothelial cells, within the sinusoids, but lack basal lamina.
- c. metabolize aged erythrocytes and they digest hemoglobin.
- d. are derived from monocytes and they destroy bacteria that enter the portal blood through the large intestine.
- e. Most of them are located in the periportal region of the liver lobule, where they are very active in phagocytosis.

3. *Serous demilunes,*

- a) produce IgA
- b) border directly on salivary intercalated ducts
- c) produce mucinogen
- d) produce lysozyme
- e) occur only in parotid glands

_____ 43. During the development of the digestive system, which statement is true?

- a) omphalocele is a persistence of the herniation of abdominal contents into the proximal part of the umbilical cord.
- b) the superior two-thirds of the adult anal canal are derived from proctodeum.
- c) the junction of the epithelium is derived from the ectoderm of the proctodeum.
- d) the inferior one-third of the anal canal is supplied with superior rectal artery.
- e) the anal membrane normally breaks down by the end of the second trimester..

_____ 44. Concerning the development of the GIT and associated structures, which statement is true?

- a) The midgut begins with 1/2 of the duodenum and ends with 1/3 of the sigmoid colon.
- b) Stomach rotates 270° counterclockwise, similar to the midgut.
- c) The middle of the duodenum would receive blood from celiac and superior mesenteric artery.
- d) The hepatocytes of liver are of mesodermal in origin, while the stroma is of endodermal origin.
- e) The exocrine portion of pancreas develops from endoderm of the gut, while the endocrine portion develops from the local mesoderm.